

# Project 1

Anita's Veg Cafe

## Context

At the beginning of 2021, Anita, a passionate home cook from Bengaluru, decided to open a small vegetarian café called Anita's Veg Café.

Her café quickly became popular among young professionals and college students who preferred affordable yet homely vegetarian meals. The café serves a small but well-loved menu:

- Paneer Butter Masala – rich and creamy, always in demand
- Veg Biryani – a flavorful rice dish, perfect for sharing
- Masala Dosa – a South Indian classic that sells throughout the day

Like any modern business owner, Anita started keeping track of her sales and customer data. After a few weeks, some of her regulars suggested she introduce a loyalty program to reward repeat customers. She agreed, and customers who joined the program started earning points for their purchases.

Now Anita wants to dig deeper into her café's data to understand:

- Which dishes are most loved
- How customer loyalty is shaping behavior
- What strategies can help her grow the business

She has asked you, as her data consultant, to help her analyze the data using SQL.

## Dataset Overview

The data is stored under the schema: `anitas_veg_cafe` with 3 tables:

1. `sales`

Captures every item ordered by each customer.

- `customer_id`
- `order_date`
- `product_id`

2. `menu`

Details of each item on the menu.

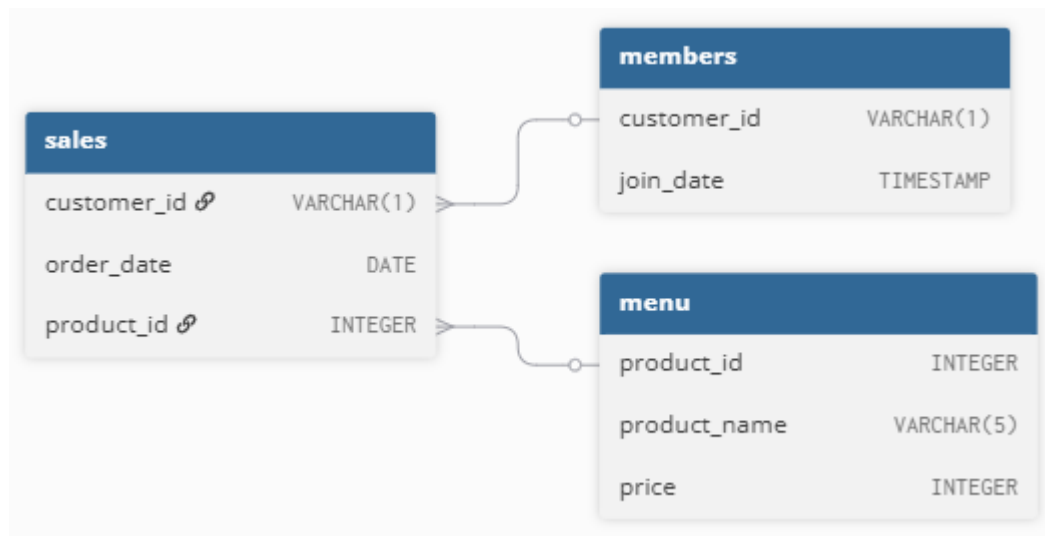
- `product_id`
- `product_name`
- `price`

3. `loyal_customers`

Customers who joined Anita's loyalty program.

- `customer_id`
- `join_date`

## Entity Relationship Diagram



## Sample Data

1. sales - The sales table captures all *customer\_id* level purchases with an corresponding *order\_date* and *product\_id* information for when and what menu items were ordered.

| customer_id | order_date | product_id |
|-------------|------------|------------|
| Aarav       | 2021-01-01 | 1          |
| Aarav       | 2021-01-01 | 2          |
| Aarav       | 2021-01-07 | 2          |
| Aarav       | 2021-01-10 | 3          |
| Aarav       | 2021-01-11 | 3          |
| Aarav       | 2021-01-11 | 3          |
| Meera       | 2021-01-01 | 2          |
| Meera       | 2021-01-02 | 2          |
| Meera       | 2021-01-04 | 1          |
| Meera       | 2021-01-11 | 1          |
| Meera       | 2021-01-16 | 3          |
| Meera       | 2021-02-01 | 3          |
| Rohan       | 2021-01-01 | 3          |
| Rohan       | 2021-01-01 | 3          |
| Rohan       | 2021-01-07 | 3          |

2. menu - The *menu* table maps the *product\_id* to the actual *product\_name* and *price* of each menu item.

| product_id | product_name         | price |
|------------|----------------------|-------|
| 1          | Paneer Butter Masala | 180   |
| 2          | Veg Biryani          | 150   |
| 3          | Masala Dosa          | 120   |

3. members - The final *members* table captures the *join\_date* when a *customer\_id* joined the beta version of Anita's loyalty program.

| customer_id | join_date  |
|-------------|------------|
| Aarav       | 2021-01-07 |
| Meera       | 2021-01-09 |

## Goal

By the end of this project, you will gain hands-on practice with joins, aggregations, window functions, case statements, and date logic — all while analyzing a real-world inspired restaurant business case.

## Analytical Questions

Help Anita answer the following questions using SQL:

1. What is the total amount each customer has spent at the café?
2. How many distinct days has each customer placed an order?
3. What was the first dish ordered by each customer?
4. Which menu item is the most popular overall?
5. What is the most frequently ordered dish for each customer?
6. After joining the loyalty program, what dish did each member first order?
7. Before joining the loyalty program, what dish did each customer order last?
8. For each member, how many items and how much did they spend *before joining*?
9. If each ₹1 = 10 points, and Paneer Butter Masala earns double points, how many points does each customer earn?
10. In their first loyalty week (starting from `join_date`), members earn double points on all items. How many points do Aarav and Meera have by the end of January?